# Integrating an Econometric Demand Model with Deep Q-Network Pricing: A Simulation Study

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***Abstract*—Retail dynamic pricing requires both demand structure and sequential adaptivity; purely econometric and purely model-free reinforcement-learning (RL) methods address complementary aspects of this problem. We describe *EG-DQN* (Econometric-Guided Deep Q-Network), a hybrid agent that incorporates Generalized Linear Model (GLM) demand signals into a Deep Q-Network through state augmentation, econometric reward shaping, and early action masking. We evaluate the design in a controlled simulation of three substitute products over 1,460 days of synthetic daily demand, with a 120-day holdout on Item 1 and comparison of fixed-price, rule-based, GLM-optimal, pure DQN, and EG-DQN policies. On the holdout window, GLM-optimal pricing and EG-DQN attain identical cumulative revenue and select the same price on the discrete grid; a one-sided Mann–Whitney test does not indicate higher daily revenue for EG-DQN relative to the GLM benchmark ($p = 0.50$). Relative to fixed-price and rule-based baselines, EG-DQN yields higher holdout revenue in this simulation (Mann–Whitney $p < 0.001$ in both comparisons). Pure DQN attains lower holdout revenue than GLM-optimal and EG-DQN under the learned policies. An ablation suggests that econometric reward shaping accounts for the principal holdout improvement over pure DQN, while the full hybrid does not exceed a reward-only configuration on test revenue. Training-episode rewards are modestly higher for EG-DQN than for pure DQN. These findings are confined to a synthetic, per-item, discrete-action environment with partial alignment between the data-generating process and the GLM specification; operational deployment is not claimed.**

***Index Terms*—Dynamic pricing, reinforcement learning, deep Q-network, generalized linear model, price elasticity, demand forecasting, simulation study, retail pricing.**

## I. Introduction

Retail managers set prices under incomplete information. When tomorrow's price is chosen, competitor reactions, seasonal shifts, and the price sensitivity of the current customer pool are not directly observable. Mistakes are costly: a price set too low forfeits revenue that cannot be recovered, and a price set too high suppresses demand immediately.

The literature has developed two complementary approaches. Econometric methods estimate price-response relationships from historical transactions, yielding interpretable parameters—in particular, own-price elasticity—that managers can audit and reconcile with demand theory [1]. Their optimisation is typically open-loop and rests on assumptions such as relatively stable demand relationships over the estimation window. Reinforcement learning (RL), by contrast, treats pricing as a sequential decision problem in which an agent learns a policy from repeated market interaction [3]. RL adapts to feedback without committing to a fixed parametric form, yet retail applications must contend with sample-efficient learning, noisy rewards, and exploration over large price sets [13], [15].

Apte et al. [13] show that tabular Q-learning can improve on static pricing in simulated retail settings. Groeneveld et al. [3] report that RL remains competitive under demand-model misspecification but can learn more slowly when demand structure is known—the regime in which econometric guidance may be most informative. When live experimentation is impractical, simulation-based evaluation offers a controlled alternative [17], [18]. We adopt that evaluation mode here; operational deployment is not claimed.

We describe *EG-DQN* (Econometric-Guided Deep Q-Network), which couples a Generalized Linear Model (GLM) demand specification with a Deep Q-Network (DQN) pricing agent. GLM-derived signals enter the agent through three mechanisms: state augmentation with the GLM-optimal price at each timestep, econometric reward shaping, and early action masking during exploration. The design is assessed in a synthetic environment with three substitute products, 1,460 days of daily demand, and a 120-day holdout on Item 1; we compare fixed-price, rule-based, GLM-optimal, pure DQN, and EG-DQN policies using holdout revenue, Mann–Whitney tests, and a three-configuration ablation.

*Contributions*

- We **describe** EG-DQN and three mechanisms for injecting GLM demand signals—optimal-price guidance, reward shaping, and masked exploration—into DQN-based retail pricing.
- We **implement** a reproducible simulation benchmark with known elasticities, five baseline strategies, holdout evaluation on Item 1, non-parametric significance testing, and ablation of econometric reward shaping.
- We **report** that GLM-optimal and EG-DQN policies coincide on holdout revenue in this simulation while exceeding naive fixed-price and rule-based strategies; econometric reward shaping aligns RL outcomes with

GLM-guided prices, and EG-DQN attains modest training-phase gains over pure DQN, without claiming holdout dominance over econometric pricing alone.

## II. Related Work

### A. Dynamic Pricing and RL

Chenavaz and Dimitrov [2] review how artificial intelligence methods have entered dynamic pricing research across operations and applied economics. Kopalle et al. [1] summarise managerial uses of dynamic pricing and the econometric questions that remain open in retail practice. On the algorithmic side, Groeneveld et al. [3] compare reinforcement learning with data-driven dynamic programming in finite-horizon markets; Razumovskiy and Karenin [4] report a related finite-horizon comparison of fitted dynamic programming and RL. Liu et al. [5] extend deep RL to joint omni-channel pricing and inventory control when demand is uncertain. Li and Zheng [6] study pricing under inventory constraints and external information within an online learning framework—a reminder that demand estimation and sequential price setting are seldom separable in applied settings.

### B. Demand Forecasting for Pricing

Pricing rules draw on demand forecasts, elasticity estimates, or both. Zhang and Meng [7] model time-varying price elasticity from retail time series, relaxing the constant-elasticity assumption embedded in many static GLM specifications. Bu et al. [8] analyse offline demand learning when sales are censored by stockouts, a common complication in retail transaction data. Ramos et al. [9] benchmark multivariate forecasters on retail sales series at the SKU level. Xia et al. [10] generate synthetic retail demand with known elasticities so that estimated parameters can be checked against ground truth; we adopt that validation logic in our simulation benchmark.

### C. Econometric Demand Models

For count demand, a log-linked GLM remains a standard specification when managers require interpretable own- and cross-price effects [7]. Safonov [12] contrasts neural demand estimators with econometric models in retail and notes that flexible machine-learning fit can underuse sparse within-product price variation that a GLM isolates. The implication for hybrid pricing is to treat econometric structure as an auditable input to learning, not as the only admissible policy.

### D. RL for Pricing

Apte et al. [13] apply tabular Q-learning to retail pricing without an explicit structural demand block in the agent. Alamdar and Seifi [14] extend deep Q-learning to joint ordering and pricing decisions. Cheung and Simchi-Levi [15] study nonstationary RL and the cost of unrestricted early exploration—one motivation for restricting actions while a demand model is still imprecise. Zheng et al. [16] combine learned demand representations with a dual-agent deep RL policy for pricing and replenishment; their demand-learning layer is the closest architectural neighbour to our GLM-guided agent. When live price experiments are costly or infeasible, offline and simulation-based evaluation protocols offer a practical alternative [17].

Prior retail pricing studies typically pursue either econometric optimisation from a fixed demand model or end-to-end RL with limited structural transparency. Few papers specify separate channels—state information, reward shaping, and action restrictions—through which econometric estimates enter an RL policy, then compare the resulting hybrid against GLM-only and pure DQN baselines on a holdout window with known elasticities. EG-DQN is intended to document that integration in simulation; we do not present it as a deployment-ready pricing system.

## III. Problem Formulation

### A. Market Structure

We consider a single retailer managing $n = 3$ substitute products over a discrete horizon of $T$ days, following multi-product retail pricing formulations [19]. At each period $t \in \{1, \dots, T\}$, the retailer observes state information and sets a price for the focal product from a finite action space. Cross-product substitution is captured in the demand system below; RL policies are estimated *per product*, with rival prices entering as inputs to the econometric model rather than as joint actions in a single multi-product MDP. The objective is a pricing policy $\pi : \mathcal{S} \to \mathcal{A}$ that maximises expected cumulative revenue:

$$\pi^* = \arg\max_{\pi} \mathbb{E}\left[\sum_{t=1}^{T} p_t \cdot D_t(p_t, s_t)\right], \tag{1}$$

where $p_t$ and $D_t$ denote the price and realised demand of the focal item on day $t$.

### B. Demand Data-Generating Process

Simulated demand follows a log-linear multiplicative structure grounded in demand theory [1], [7]:

$$D_{it} = B_i(t) \cdot S_i(t) \cdot \prod_{j=1}^{n} \left(\frac{p_{jt}}{\bar{p}_j}\right)^{\varepsilon_{ij}} + \eta_{it}, \tag{2}$$

where $B_i(t)$ is a time-varying base-demand component (trend), $S_i(t)$ is a multiplicative seasonal index, $\varepsilon_{ij}$ is the cross-price elasticity of product $i$ with respect to product $j$'s price (own-price elasticity when $i = j$), $\bar{p}_j$ is a reference price, and $\eta_{it} \sim \mathcal{N}(0, \sigma_i^2)$ is i.i.d. Gaussian noise. Because $\log \mathbb{E}[D_{it}]$ is linear in $\log p_{jt}$, the Poisson GLM with log-link used later is *approximately* aligned with the simulator; it is not treated as a correctly specified model for the noise term or for all trend–seasonal components.

### C. Pricing as a Markov Decision Process

The focal product's pricing problem is written as an MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$.

**State space $\mathcal{S}$:** $s_t \in \mathbb{R}^5$ comprises: (i) previous-period price; (ii) previous-period observed demand; (iii) day-of-year sine encoding annual seasonality; (iv) normalised time index $t/T$;

and (v) the GLM-optimal price $p^*_{\mathrm{GLM},t}$. Component (v) is the econometric signal that distinguishes EG-DQN from pure DQN.

**Action space $\mathcal{A}$:** A discrete grid of $K = 20$ prices on $[p_{\min}, p_{\max}]$. During early training, EG-DQN restricts feasible actions to a neighbourhood of $p^*_{\mathrm{GLM},t}$ (action masking).

**Reward — Pure DQN:**

$$r_t = p_t \cdot \max\bigl(0,\ \hat{D}_t(p_t, s_t)\bigr), \tag{3}$$

where $\hat{D}_t$ is the GLM demand forecast used as a revenue proxy.

**Reward — EG-DQN:**

$$r_t^{\mathrm{EG}} = r_t \ +\ \lambda \cdot \exp\!\left(-\frac{|p_t - p^*_{\mathrm{GLM},t}|}{\sigma_p}\right), \tag{4}$$

with econometric weight $\lambda$ and price scale $\sigma_p$. The bonus attracts the agent toward GLM-endorsed prices without fixing the selected price.

**Discount factor:** $\gamma = 0.99$, consistent with daily pricing over finite training episodes.

## IV. Econometric Demand Model

### A. Specification

We estimate a separate Poisson GLM with log-link for each product $i$. Count demand is natural for daily units sold; the log-link yields interpretable own- and cross-price elasticities of the form common in retail pricing research [1], [7]. For product $i$,

$$\log\bigl(\mathbb{E}[D_{it}]\bigr) = \alpha_0 + \underbrace{\alpha_1 \log p_{it}}_{\text{own elasticity}} + \sum_{j\neq i} \beta_j \log p_{jt} + \gamma_1\, \mathrm{trend}_t + \gamma_2 \sin\!\left(\tfrac{2\pi\, \mathrm{day}_t}{365}\right) + \gamma_3\, \mathrm{weekend}_t, \tag{5}$$

where $\alpha_1$ is the own-price elasticity, $\beta_j$ are cross-price elasticities, and $\gamma_k$ capture trend, annual seasonality, and weekend effects. Coefficients are estimated by maximum likelihood on the training window only; HC3-robust standard errors are reported. Holdout forecasts and GLM-optimal prices use this fixed fit—we do not refit online on the 120-day test window. As noted in Section III, the log-linear DGP is only approximately aligned with (5); time-varying elasticity is outside the present specification [7].

### B. Derived Econometric Signals

Three GLM outputs feed the hybrid agent [12]:

1) **Estimated elasticity** $\hat{\varepsilon}_i = \hat{\alpha}_1$. In simulation we compare $\hat{\varepsilon}_i$ to the known DGP own-price elasticity as an *elasticity recovery* check, not as a claim of structural identification.
2) **GLM-optimal price** $p^*_{\mathrm{GLM},t}$. For each day $t$ and covariate vector $\mathbf{x}_t$, we evaluate expected revenue $R(p) = p \cdot \hat{D}_i(p, \mathbf{x}_t)$ on the discrete price grid and set $p^*_{\mathrm{GLM},t} = \arg\max_p R(p)$. This quantity enters the EG-DQN state, the early action mask, and the econometric reward bonus.
3) **Forecast demand** $\hat{D}_{it}$. The fitted mean demand at the chosen price serves as a revenue proxy in (3), limiting the effect of day-to-day demand noise on the learning signal [8].

Estimation accuracy and holdout forecast metrics are deferred to the experimental results; this section defines only the signals passed to the DQN.

## V. Reinforcement Learning Framework

### A. Network Architecture

Each product is assigned its own Deep Q-Network agent. The policy network $Q(s, a; \theta)$ maps the five-dimensional state vector defined in Section III to a Q-value for each of $K = 20$ discrete candidate prices. The network has two hidden layers of 128 units with ReLU activations and Glorot initialisation; parameters are updated in NumPy without a deep-learning framework dependency, which keeps the simulation reproducible on standard hardware. Retail Q-learning studies motivate discrete-action pricing agents of this form [13], [14]. Temporal-difference targets follow the standard DQN backup with a separate target network $\theta^-$; the regression loss is Huber with $\delta = 1$ rather than squared error, which limits the influence of occasional high-revenue transitions:

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathcal{B}}\Bigl[\ell_\delta\Bigl(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta)\Bigr)\Bigr]. \tag{6}$$

### B. Training Procedure

Training proceeds for 300 episodes per product. At each environment step the agent stores transitions $(s_t, a_t, r_t, s_{t+1})$ in a replay buffer of capacity 10,000; a minibatch of 64 transitions is drawn uniformly for each gradient update. The target network weights are copied from the online network every 15 episodes. Action selection is $\varepsilon$-greedy with $\varepsilon$ decaying from 1.0 to 0.05 over training; gradient norms are clipped at $\ell_2 = 1.0$. The learning rate is $10^{-3}$. Pure DQN uses the reward in (3); EG-DQN variants and econometric hyperparameters ($\lambda$, masking horizon) are specified in the next subsection.

### C. EG-DQN Integration Mechanisms

EG-DQN extends the base DQN with three channels through which the fixed GLM fit from Section IV enters learning. Pure DQN omits all three; the ablation in the results section toggles reward shaping while holding the remaining protocol fixed.

- **State augmentation.** The state vector includes $p^*_{\mathrm{GLM},t}$ at every step (component (v) in Section III), giving the agent an explicit econometric price benchmark alongside lagged market variables.
- **Reward shaping.** Training rewards follow (4) with $\lambda = 30.0$ and $\sigma_p$ set to the standard deviation of the discrete price grid. The exponential bonus is largest when the selected price coincides with $p^*_{\mathrm{GLM},t}$ and decays smoothly with price distance.
- **Early action masking.** For episodes $e \leq 30$, feasible actions are restricted to grid points within $\pm\sigma_p$ of $p^*_{\mathrm{GLM},t}$, limiting exploratory prices while the replay buffer is still sparse. The full grid is available from episode 31 onward. Restricting early exploration in nonstationary settings can reduce the cost of uninformed trial prices [15].

Algorithm 1 summarises one training episode; the outer loop runs $E = 300$ episodes per product with the hyperparameters in Section V.

**Algorithm 1** EG-DQN Training (single episode)

**Require:** Fitted GLM $\mathcal{G}$, environment $\mathcal{M}$, replay buffer $\mathcal{B}$, weights $\theta$, target $\theta^{-}$, exploration rate $\varepsilon$

```
s_0 ← RESET(); compute p*_{GLM,0} from G
for step t = 0, 1, ... until episode end do
    if current episode e ≤ 30 then A_t ← {p ∈ grid : |p − p*_{GLM,t}| ≤ σ_p}
    else A_t ← full price grid
    end if
    a_t ← ε-greedy(Q(s_t, ·; θ), A_t)
    Execute a_t; observe r_t^EG from (4)
    s_{t+1} ← next state; store (s_t, a_t, r_t^EG, s_{t+1}) in B
    if |B| ≥ 64 then
        Sample minibatch from B; update θ via (6) with gradient clip
    end if
    s_t ← s_{t+1}
end for
Decay ε; if e mod 15 = 0 then θ^- ← θ
```

## VI. Simulation Environment

### A. Design and Calibration

We evaluate EG-DQN in a controlled synthetic market rather than a live retail test. Simulation allows known ground-truth elasticities, a fixed holdout protocol, and reproducible comparison of pricing policies under identical demand trajectories—an evaluation mode used for offline and retail RL studies when field experimentation is costly or infeasible [10], [11], [17], [18]. We do not treat the synthetic setting as a substitute for real-store validation; it is a design and benchmark study.

The simulator generates daily demand for $n = 3$ substitute products over $T =$ 1,460 days (1 January 2021 – 30 December 2024). The final 120 days form an out-of-sample holdout; the main revenue comparison in the results section uses Item 1 on that window. Demand parameters are calibrated to statistical properties typical of large retail benchmarks (M5 / Walmart-style series and related synthetic generators [10]): prices on a \$6–\$18 range, demand coefficient of variation of approximately 0.25, and clear annual seasonality. All stochastic draws and training runs use seed 42.

Environment components are as follows.

- **Trend.** Base demand grows linearly, $B_i(t) = \text{base}_i \cdot (1 + g_i \cdot t/T)$, with base levels (50, 40, 60) units/day for Items 1–3.
- **Seasonality.** An annual sinusoid (amplitudes 12, 8, 15) plus an 8% weekend uplift.
- **Price paths.** Step-function list prices with roughly 60 discrete changes per product over four years, mimicking intermittent promotions.
- **Noise.** $\eta_{it} \sim \mathcal{N}(0, \sigma_i^2)$ with $\sigma = (4.0, 3.5, 5.0)$.

Table I reports the cross-price elasticity matrix used in (2). All own-price elasticities satisfy $|\varepsilon_{ii}| > 1$, so demand is elastic at the reference prices and a revenue-maximising price exists in the continuous problem. As noted in Section III, the DGP is only approximately aligned with the Poisson log-link GLM; elasticities recovered by the GLM are therefore a sanity check against known simulator parameters, not external validation.

TABLE I: Cross-price elasticity matrix ($\varepsilon_{ij}$) used in the simulator. Diagonal entries are own-price elasticities; off-diagonal entries are substitution (positive) or complementary (negative) effects.

| | **Item 1** | **Item 2** | **Item 3** |
|---|---|---|---|
| **Item 1** | −1.50 | +0.35 | −0.15 |
| **Item 2** | +0.25 | −1.80 | +0.10 |
| **Item 3** | −0.20 | +0.15 | −1.20 |

### B. Hyperparameters

Table II lists the simulation and training settings used for all reported runs. The discrete action set comprises $K = 20$ prices spanning \$6.0 to \$16.0, matching the grid used in the experimental notebook. Learning parameters follow a standard DQN recipe; the econometric penalty weight is $\lambda = 30.0$ with $\sigma_p$ equal to the standard deviation of the price grid. One DQN (or EG-DQN) agent is trained *per product*; there is no joint multi-product action space in this study.

TABLE II: Simulation and training hyperparameters

| **Parameter** | **Value** | **Notes** |
|---|---|---|
| State dimension | 5 | See Section III |
| Action count $K$ | 20 | Grid \$6.0–\$16.0 |
| Training episodes | 300 | Per product |
| Batch size | 64 | Replay minibatch |
| Replay buffer | 10,000 | FIFO experience store |
| Discount $\gamma$ | 0.99 | Daily pricing horizon |
| Learning rate $\alpha$ | $10^{-3}$ | Gradient updates |
| Target sync | 15 episodes | Hard copy $\theta^{-} \leftarrow \theta$ |
| $\varepsilon$ start/end | 1.0 / 0.05 | Linear decay |
| Masking window | episodes 1–30 | EG-DQN only |
| $\lambda$ (EG-DQN) | 30.0 | Reward-shaping weight |
| Holdout length | 120 days | Item 1 revenue focus |
| Random seed | 42 | Full pipeline |

### C. Baseline Strategies

Five pricing strategies are compared on the same simulated trajectories and the same 120-day holdout for Item 1.

- **Fixed-Price (FP).** Charge the reference price $\bar{p}$ every day. This is a naive non-adaptive floor.
- **Rule-Based (RB).** Set price to $1.15\times$ the trailing 7-day mean of observed prices. The rule is a simple retailer heuristic, not an optimised policy.
- **GLM-Optimal (GLM).** Each day, charge $p^{*}_{\text{GLM},t}$ from the fitted Poisson model (Section IV). This is the pure econometric benchmark on the discrete grid.
- **Pure DQN.** The network and training procedure of Section V without state augmentation of $p^{*}_{\text{GLM},t}$, without reward shaping, and without action masking.
- **EG-DQN.** The full hybrid: all three integration mechanisms of Section V.

Holdout evaluation uses realised simulator demand under each policy's chosen prices. Statistical comparisons of daily revenue series follow in the results section; the present section only fixes the protocol.

## VII. Experiments and Results

### A. Demand Estimation Accuracy

We first check that the fitted Poisson GLM recovers usable demand structure in the calibrated simulator before pricing policies are compared. Figure 1 summarises the three-product series: step-function prices, seasonal demand, log–log price–demand scatters, and marginal demand distributions over the full horizon.

Table III compares estimated own-price elasticities with the ground-truth parameters of (2). Relative absolute biases are 1.2%, 1.7%, and 3.7% for Items 1–3 respectively (largest on Item 3). We therefore report elasticity recovery *within about* 3.7% in this run, rather than a tighter uniform bound. Cross-price signs match the substitution structure in Table I. Figure 2 overlays GLM and true demand curves against price: slopes agree closely with the simulator, while fitted *levels* sit above observed demand—a level misfit we treat as a modelling caveat rather than as formal structural identification.

TABLE III: GLM own-price elasticity estimates versus simulator truth (HC3-robust standard errors). Relative bias is $|\hat{\varepsilon} - \varepsilon|/|\varepsilon|$.

| **Product** | **True $\varepsilon$** | **Est. $\hat{\varepsilon}$** | **Bias** | **Rel. bias** |
|---|---|---|---|---|
| Item 1 | $-1.500$ | $-1.519$ | $-0.019$ | 1.2% |
| Item 2 | $-1.800$ | $-1.770$ | $+0.030$ | 1.7% |
| Item 3 | $-1.200$ | $-1.245$ | $-0.045$ | 3.7% |

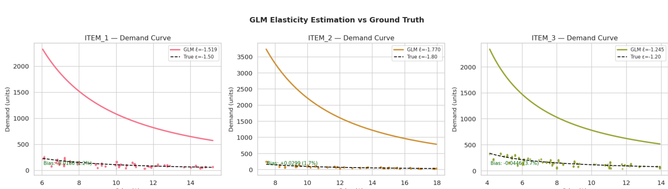


Fig. 2: GLM demand curves versus ground-truth DGP curves for Items 1–3. Annotated relative biases refer to own-price elasticity recovery (1.2%, 1.7%, 3.7%); curve *levels* are not claimed to match.

Table IV reports in-sample fit and 120-day holdout forecast accuracy for Item 1, the product used in the revenue comparison. Holdout MAPE is 5.20% and $R^2$ is 0.965. Figure 3 shows fitted and forecast paths against realised demand; Panel B residuals on the holdout window have mean approximately $-4.57$ units/day, indicating a slight systematic over-forecast even though percentage error remains modest.

TABLE IV: GLM forecast performance for Item 1 (Poisson log-link). Train covers the pre-holdout window; test is the final 120 days.

| **Metric** | **Train** | **Test (120-day)** |
|---|---|---|
| MAE (units/day) | 4.80 | 5.81 |
| RMSE | 6.17 | 7.07 |
| MAPE (%) | 5.22 | 5.20 |
| $R^2$ | 0.983 | 0.965 |

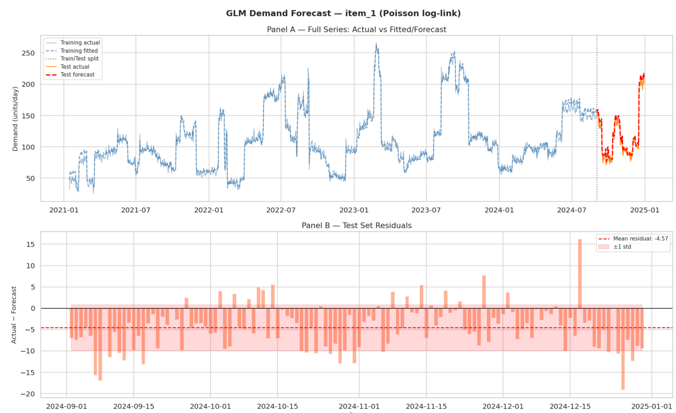


Fig. 3: GLM demand fit and 120-day holdout forecast for Item 1 (Poisson log-link). Panel A: training actual/fitted and test actual/forecast. Panel B: test residuals (actual − forecast); mean residual $\approx -4.57$ units/day.

## References


[1] P. K. Kopalle, K. Pauwels, L. Y. Akella, and M. Gangwar, "Dynamic pricing: Definition, implications for managers, and future research directions," *Journal of Retailing*, vol. 99, no. 4, pp. 580–593, 2023. https://doi.org/10.1016/j.jretai.2023.11.003

[2] R. Y. Chenavaz and S. Dimitrov, "Artificial intelligence and dynamic pricing: A systematic literature review," *J. Appl. Econ.*, vol. 28, no. 1, p. 2466140, 2025. https://doi.org/10.1080/15140326.2025.2466140

[3] J. Groeneveld *et al.*, "Reinforcement learning versus data-driven dynamic programming for finite horizon dynamic pricing markets," *Journal of Revenue and Pricing Management*, 2025. https://doi.org/10.1057/s41272-025-00519-8

[4] L. Razumovskiy and N. Karenin, "A comparative study of dynamic programming and reinforcement learning in finite-horizon dynamic pricing," *arXiv:2604.14059*, 2025.

[5] S. Liu *et al.*, "Data-driven dynamic pricing and inventory management of an omni-channel retailer in an uncertain demand environment," *Expert Syst. Appl.*, vol. 244, p. 122948, 2024. https://doi.org/10.1016/j.eswa.2023.122948

[6] X. Li and Z. Zheng, "Dynamic pricing with external information and inventory constraint," *Manag. Sci.*, 2023. https://doi.org/10.1287/mnsc.2023.4963

[7] W. Zhang and Y. Meng, "Dynamic modeling and forecasting of price elasticity based on time series analysis and machine learning," *SAGE Open Economics*, 2025. https://doi.org/10.1177/14727978251338001

[8] J. Bu, D. Simchi-Levi, and L. Wang, "Offline pricing and demand learning with censored data," *Management Science*, vol. 69, no. 2, pp. 885–903, 2023.

[9] P. Ramos *et al.*, "Retail demand forecasting: A comparative study for multivariate time series," *arXiv:2308.11939*, 2023.

[10] Y. Xia, A. Arian, S. Narayanamoorthy, and J. Mabry, "RetailSynth: Synthetic data generation for retail AI systems evaluation," *arXiv:2312.14095*, 2023.

[11] Y. Xia *et al.*, "Simulation-based benchmarking of reinforcement learning agents for personalized retail promotions," *arXiv:2405.10469*, 2024.

[12] K. Safonov, "Neural network approach to demand estimation and dynamic pricing in retail," *arXiv:2412.00920*, 2024.

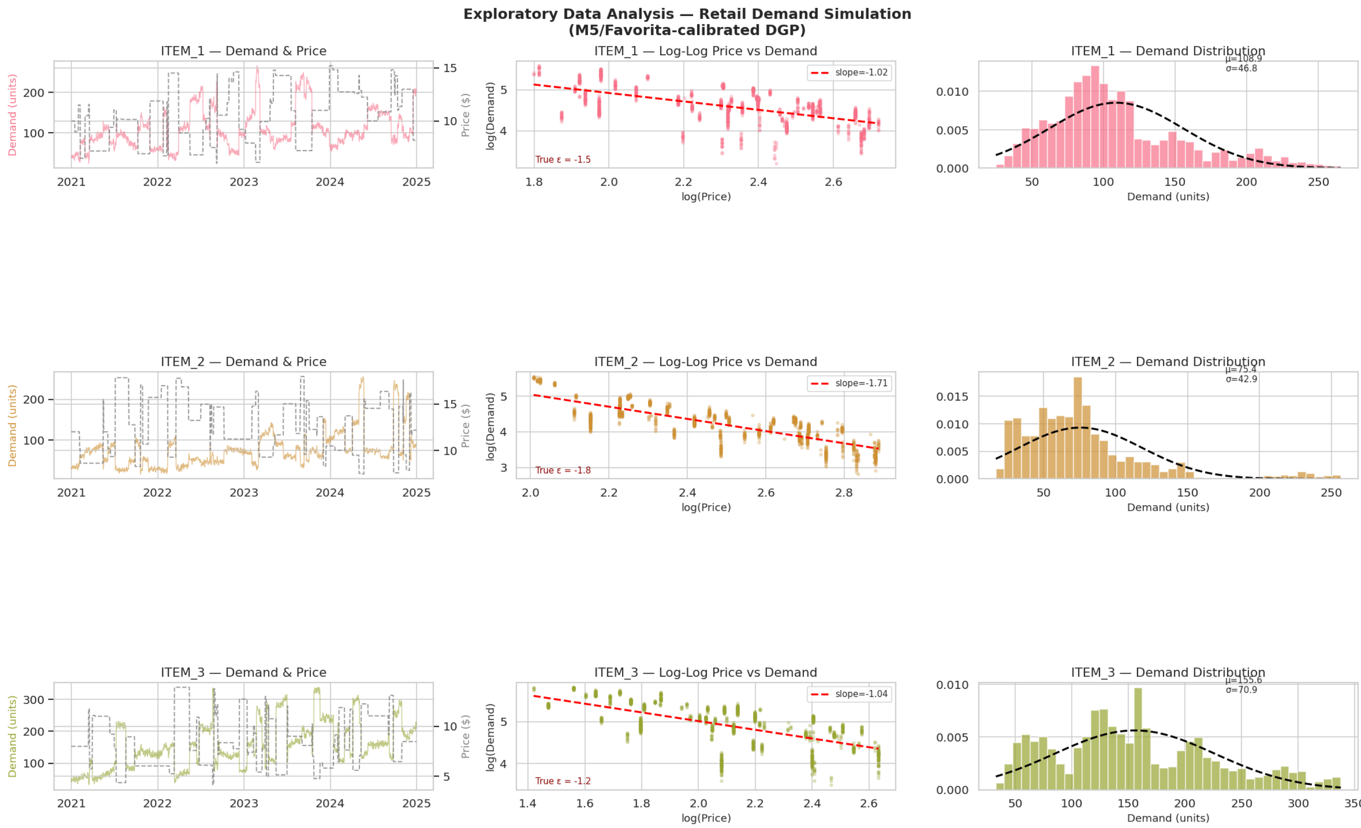


Fig. 1: Exploratory view of the synthetic retail series (Items 1–3). Left: daily demand (solid) and list price (dashed). Centre: log–log price–demand scatter with OLS slope. Right: demand histogram with a normal overlay. The horizon spans 2021–2024 under the M5/Favorita-calibrated DGP.


[13] M. Apte, K. Kale, P. Datar, and P. R. Deshmukh, "Dynamic retail pricing via Q-Learning — A reinforcement learning framework for enhanced revenue management," *arXiv:2411.18261*, 2024.

[14] P. F. Alamdar and A. Seifi, "A deep Q-learning approach to optimize ordering and dynamic pricing decisions," *International Journal of Production Economics*, vol. 269, p. 109154, 2024.

[15] W. C. Cheung and D. Simchi-Levi, "Nonstationary reinforcement learning: The blessing of (more) optimism," *Manag. Sci.*, 2023. https://doi.org/10.1287/mnsc.2023.4704

[16] Y. Zheng *et al.*, "Dual-agent deep reinforcement learning for dynamic pricing and replenishment," *arXiv:2410.21109*, 2024.

[17] M. Nambiar *et al.*, "Offline deep reinforcement learning for dynamic pricing of consumer credit," *arXiv:2203.03003*, 2022.

[18] J. Chen, Y. Xu, and P. Yu, "A reinforcement learning approach for hotel revenue management with evidence from field experiments," *Journal of Operations Management*, vol. 69, no. 7, pp. 1176–1201, 2023.

[19] J. H. Rios and J. R. Vera, "Dynamic pricing and inventory control for multiple products in a retail chain," *Computers & Industrial Engineering*, vol. 177, p. 109065, 2023.